

Minute-05

Date: 2026-07-06

PROJECT NAME: Plagiarism Detector

SUPERVISOR: Er. Prativa Nyaupane

TEAM LEADER (TL): Neon Neupane

MEMBER(M1): Bishal Acharya

MEMBER(M2): Nabin Giri

PROGRESS CHECK: (Tick ✓ in appropriate box)

AGENDA:

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|----|----|----|----|----|----|----|----|-----|----|----|
| 5 | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 | 50 | 55 |
| 60 | 65 | 70 | 75 | 80 | 85 | 90 | 95 | 100 | | |
- Demonstration of the text extraction service and of the OCR path
 - Review of the preprocessing utilities and of the matrix component
 - Discussion on the training of the model
 - Sprint planning for the similarity engines

DISCUSSION:

The extraction service was demonstrated. Text is read correctly from PDF, DOCX and TXT files, and a scanned image is read through Tesseract, although the output becomes poor when the scan is of low quality. A PDF that has no text layer is now detected and sent to OCR automatically instead of returning an empty result. The preprocessing service was shown next; it lowercases the text, removes the extra spacing, drops the stopwords for the baseline engine and splits the text into sentences, which are the unit used for highlighting later. The similarity matrix component was shown with sample data and displays a grid of document to document scores with colour coded cells. The training work was discussed in detail. The notebook that fine tunes Sentence-BERT on the paraphrase dataset has been drafted and a training script that downloads the dataset and saves the model in one command has been committed, so that the training can be repeated by anyone during the evaluation. The supervisor asked the team to keep the notebook output in the repository as evidence of

the training, and to record the accuracy of the baseline before the trained model replaces it.

INDIVIDUAL ACHIEVEMENT (LAST SPRINT):

<u>TEAM LEADER</u>	<u>MEMBER 1</u>	<u>MEMBER 2</u>
Implemented the text extraction service with Tesseract OCR for images and for scanned PDF files, wrote the preprocessing utilities for cleaning, tokenizing and sentence splitting, drafted the training notebook and committed the one command training script for the model.	Built the similarity matrix table component with colour coded cells for the score ranges and with a legend, and tested it using sample response data prepared from the frozen API contract.	Tested the extraction service on every sample document including the scanned pages, recorded the accuracy of the OCR on clean and on poor scans, and reported the files where the extracted text needed correction.

RESPONSIBILITIES (COMING SPRINT):

<u>TEAM LEADER</u>	<u>MEMBER 1</u>	<u>MEMBER 2</u>
Fine tune the model on the Quora corpus, implement the TF-IDF baseline and the trained model as two selectable engines, and test the scraping approach in a notebook before writing the service.	Build the component that shows the text of a document with the copied sentences marked inside it.	Build the component that lists the individual sentence matches with their scores, and continue testing with the sample documents.

SIGNATURE:

MEMBER 1

MEMBER 2

TEAM LEADER

SUPERVISOR

NOTE:

Photo (all members & supervisor included) must be attached with this minute during submission.

To be filled by Supervisor:

INDIVIDUAL MARKING (5):

<u>TEAM LEADER</u>	<u>MEMBER 1</u>	<u>MEMBER 2</u>